

Sales Performance & Business Intelligence Dashboard

Portfolio Project — Data Analytics / Data Science

Prepared as a practical post-certificate portfolio project

Executive Summary

This project demonstrates an end-to-end business analytics workflow using a synthetic retail dataset. The workflow follows the same broad portfolio logic shown in the supplied IBM-style capstone report: data preparation, exploratory analysis, SQL analysis, interactive dashboard design, results, and recommendations. The dataset and results in this project are newly generated for portfolio practice.

Methodology

1. Data generation and quality checks
2. Data wrangling and derived metrics
3. Exploratory data analysis
4. SQL business questions
5. KPI and dashboard design
6. Business interpretation and recommendations

1. Introduction

The objective is to turn transaction-level retail data into actionable business intelligence. The dashboard is designed for a manager who needs a concise view of revenue, profitability, regional performance, product performance, and trends.

2. Data Description

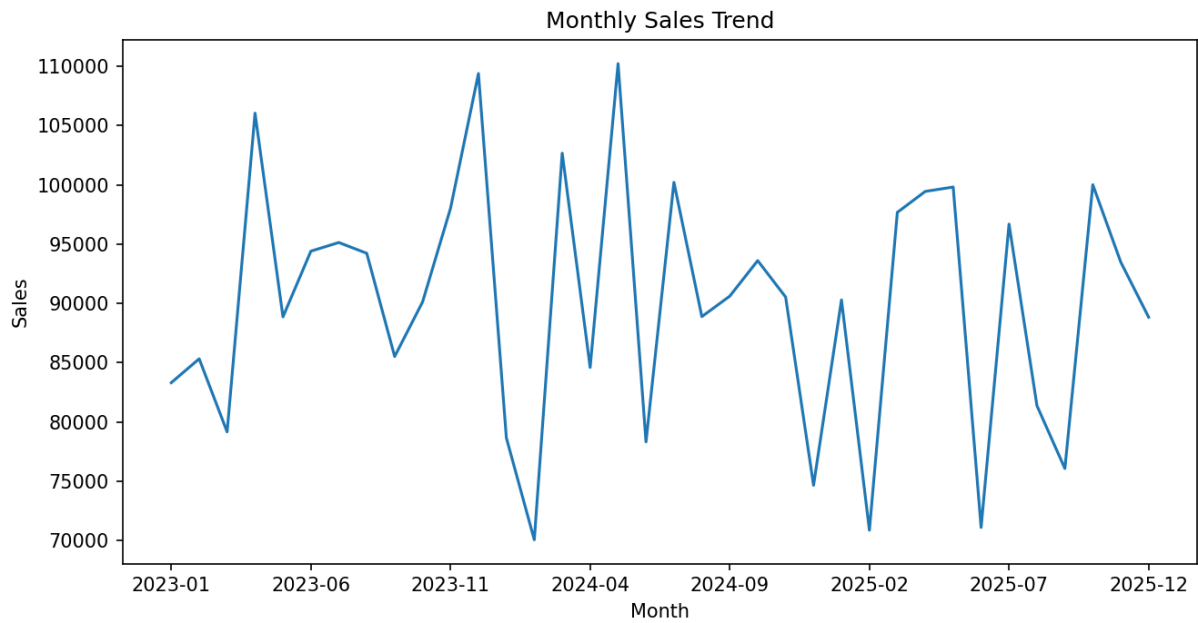
The synthetic dataset contains 6,000 orders from 2023–2025. Each row contains an order identifier, date, region, customer segment, product category, sub-category, product, quantity, unit price, discount, sales, profit, and customer identifier.

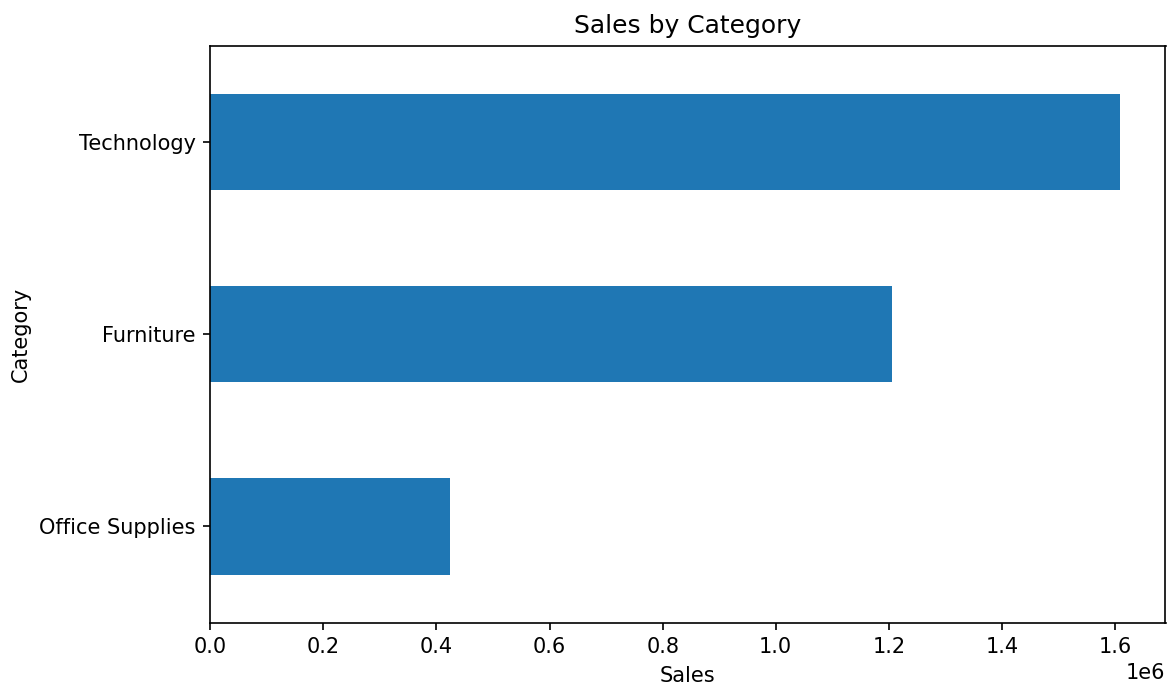
3. Key Performance Indicators

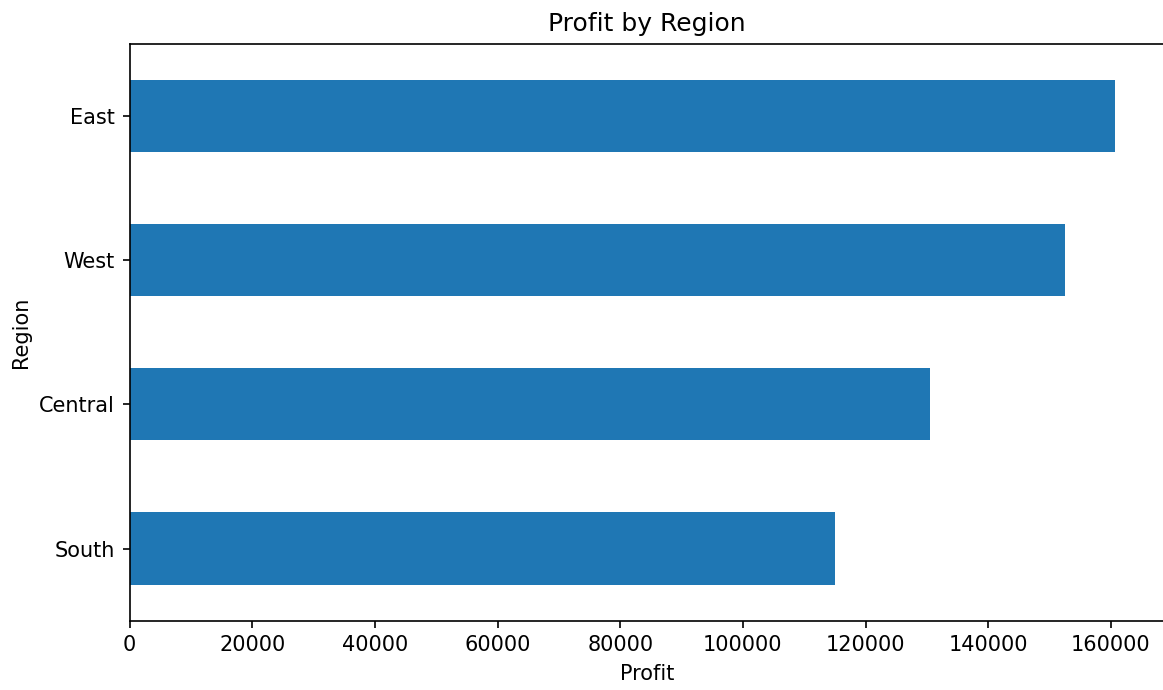
Metric	Value
Total Sales	\$3,238,406.35
Total Profit	\$558,776.64
Profit Margin	17.3%
Orders	6,000
Customers	1,191
Average Order Value	\$539.73

4. Exploratory Data Analysis

The analysis examines monthly sales, category performance, regional profitability, product performance, and discount behavior. The visuals below are generated directly from the project dataset.







5. SQL Analysis

SQL queries are included for KPI calculation, regional and category summaries, monthly trends, top products, discount analysis, and window-function ranking. The full query set is provided in `sales_analysis.sql`.

6. Dashboard Design

The recommended Power BI dashboard uses KPI cards at the top, a monthly trend chart, category and regional comparisons, a top-products view, and slicers for date, region, segment, and category. This creates an executive overview while preserving the ability to drill into detail.

7. Business Findings

Technology is the largest sales category in this synthetic dataset, followed by the other categories. Regional performance differs across both sales and profit, so the dashboard should show both measures rather than sales alone. Monthly analysis provides a time-based view for identifying growth and weaker periods. Discount analysis should be monitored because higher discounts can increase sales volume while reducing margin.

8. Recommendations

- Monitor profit margin alongside revenue.
- Investigate products with high sales but weak profit.
- Compare regional performance using both sales and profit.
- Use monthly trends to support inventory and campaign planning.
- Review discount levels and their relationship with profitability.
- Add customer-level retention metrics in a future version.

9. Portfolio / GitHub Checklist

- ✓ Dataset included
- ✓ Python analysis script
- ✓ SQL analysis
- ✓ EDA visualizations
- ✓ KPI definitions
- ✓ Power BI dashboard specification
- ✓ Business findings
- ✓ Recommendations
- ✓ README documentation

Note: This report is inspired by the organization and analytical breadth of the supplied capstone example, but all project data, results, and written analysis here are original and generated for this portfolio project.